

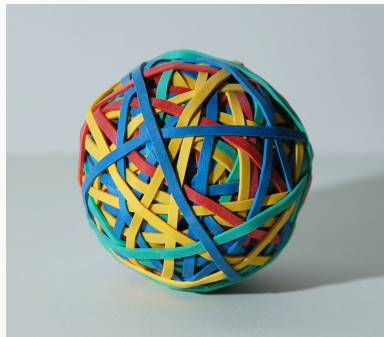
Dynamic finite element analysis of structures

A mathematical perspective

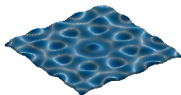
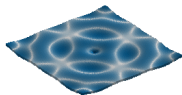
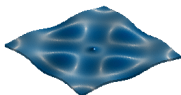
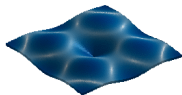
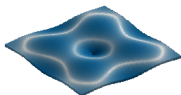
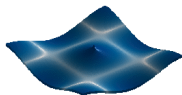
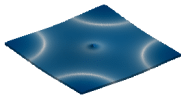
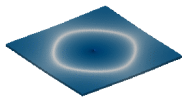
Stefano Burzio

@ Flexlab

27.01.2026



Strong and weak form of elastodynamics



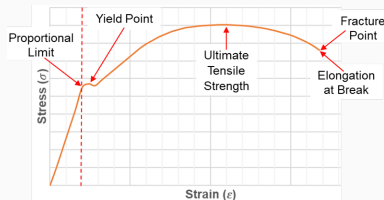
Linear: infinitesimally small deformations relative to the solid's size.

Elasticity: deformable material that returns to its original shape and size when the forces causing the deformation are removed.

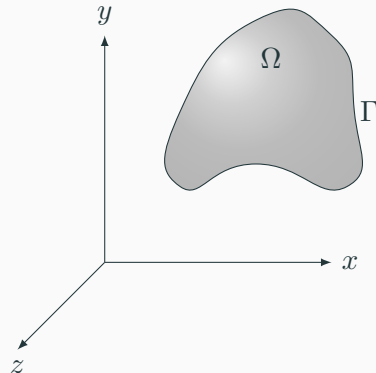
Dynamics: study bodies in motion under the influence of time-dependent loads applied to them.

Hypothesis and notations

- Inertial orthonormal reference frame $O(x, y, z)$. Sometimes will use $O(x_1, x_2, x_3)$.
- Continuous three-dimensional deformable finite (compact) **body** Ω with surface Γ .
- **Material**: continuous and homogenous.
- **Material properties**: $\mathbf{C}$ and ρ .
- **Deformations**: small (proportional to stress).



Credit: *Fidelis*



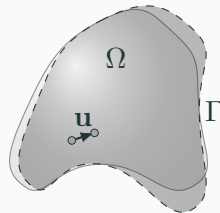
Undeformed and deformed solid

- **Unknown:** displacement $\mathbf{u} \in C^2(\bar{\Omega} \times [0, T], \mathbb{R}^3)$

Undeformed solid



Deformed solid



Loads, boundary conditions and initial conditions

■ Boundary:

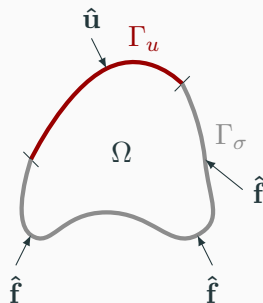
$$\Gamma = \Gamma_u \cup \Gamma_\sigma$$

■ Loads:

- Acting loads on the body: $\mathbf{f}$.
- Boundary conditions:
 - ▶ prescribed displacements $\hat{\mathbf{u}}$ on the boundary Γ_u ,
 - ▶ prescribed loads $\hat{\mathbf{f}}$ on the boundary on Γ_σ .

■ Initial conditions: at $t = 0$

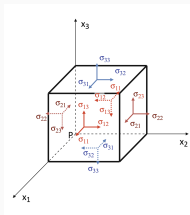
- initial displacement $\mathbf{u}_0$
- initial velocity $\mathbf{v}_0$.



Stress and strain tensors

Stress tensor

$$\boldsymbol{\sigma}(\mathbf{x}, t) = \begin{bmatrix} \sigma_{11}(\mathbf{x}, t) \\ \sigma_{22}(\mathbf{x}, t) \\ \sigma_{33}(\mathbf{x}, t) \\ \sigma_{23}(\mathbf{x}, t) \\ \sigma_{31}(\mathbf{x}, t) \\ \sigma_{12}(\mathbf{x}, t) \end{bmatrix}$$



Credit: Neto et al.

Strain tensor

$$\boldsymbol{\varepsilon}(\mathbf{x}, t) = \begin{bmatrix} \varepsilon_{11}(\mathbf{x}, t) \\ \varepsilon_{22}(\mathbf{x}, t) \\ \varepsilon_{33}(\mathbf{x}, t) \\ 2\varepsilon_{23}(\mathbf{x}, t) \\ 2\varepsilon_{31}(\mathbf{x}, t) \\ 2\varepsilon_{12}(\mathbf{x}, t) \end{bmatrix}$$

Generalised Hooke's law

The constitutive stress-strain relationship of linear elasticity is given by the Hooke's law:

$$\boldsymbol{\sigma}(\mathbf{x}, t) = \mathbf{C} \boldsymbol{\varepsilon}(\mathbf{x}, t)$$

or equivalently

$$\boldsymbol{\varepsilon}(\mathbf{x}, t) = \mathbf{S} \boldsymbol{\sigma}(\mathbf{x}, t)$$

where $\mathbf{C}$ is called the *stiffness material matrix* and $\mathbf{S} = \mathbf{C}^{-1}$.

Materials classification

Anisotropic

$$\mathbf{S} = \begin{bmatrix} s_{11} & s_{12} & s_{13} & s_{14} & s_{15} & s_{16} \\ s_{21} & s_{22} & s_{23} & s_{24} & s_{25} & s_{26} \\ s_{31} & s_{32} & s_{33} & s_{34} & s_{35} & s_{36} \\ s_{41} & s_{42} & s_{43} & s_{44} & s_{45} & s_{46} \\ s_{51} & s_{52} & s_{53} & s_{54} & s_{55} & s_{56} \\ s_{61} & s_{62} & s_{63} & s_{64} & s_{65} & s_{66} \end{bmatrix}$$

- have properties that vary with direction and do not have symmetry.
- 21 independent constants (symmetry).

Orthotropic

$$\mathbf{S} = \begin{bmatrix} \frac{1}{E_1} & -\frac{\nu_{21}}{E_2} & -\frac{\nu_{31}}{E_3} & 0 & 0 & 0 \\ -\frac{\nu_{12}}{E_1} & \frac{1}{E_2} & -\frac{\nu_{32}}{E_3} & 0 & 0 & 0 \\ -\frac{\nu_{13}}{E_1} & -\frac{\nu_{23}}{E_2} & \frac{1}{E_3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{23}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G_{31}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{12}} \end{bmatrix}$$

- Three planes of symmetry.
- E_i is the Young's modulus.
- G_{ij} is the shear modulus.
- ν_{ij} is the Poisson's ratio.
- 6 independent constants since:

$$\frac{\nu_{12}}{E_1} = \frac{\nu_{21}}{E_2}, \quad \frac{\nu_{23}}{E_2} = \frac{\nu_{32}}{E_3}, \quad \frac{\nu_{31}}{E_3} = \frac{\nu_{13}}{E_1}$$

Isotropic

$$\mathbf{S} = \begin{bmatrix} 1/E & -\nu/E & -\nu/E & 0 & 0 & 0 \\ -\nu/E & 1/E & -\nu/E & 0 & 0 & 0 \\ -\nu/E & -\nu/E & 1/E & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/G & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/G & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/G \end{bmatrix}$$

- Identical properties in all directions.
- E is the Young's modulus, G is the shear modulus, and ν is the Poisson's ratio.
- Only 2 independent constants since

$$E = 2G(1 + \nu)$$

Strain-displacement relationship

- Strain is obtained by derivatives of displacements for small deformations.
- The strain-displacement relation can be written in the *three equivalent forms* as follows:

Kinematics

a) $\boldsymbol{\varepsilon}(\mathbf{x}, t) = \nabla \mathbf{u}(\mathbf{x}, t)$

b)
$$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{23} \\ 2\varepsilon_{31} \\ 2\varepsilon_{12} \end{bmatrix} = \begin{bmatrix} \partial_x & 0 & 0 \\ 0 & \partial_y & 0 \\ 0 & 0 & \partial_z \\ 0 & \partial_z & \partial_y \\ \partial_z & 0 & \partial_x \\ \partial_y & \partial_x & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

c) $\varepsilon_{ii} = \partial_{x_i} u_i$ and $2\varepsilon_{ij} = \partial_{x_i} u_j + \partial_{x_j} u_i$

Differential equations of motion a.k.a equilibrium equations

- The equilibrium equation of elastodynamics in matrix form is:

$$\nabla^T \boldsymbol{\sigma}(\mathbf{x}, t) + \mathbf{f}(\mathbf{x}, t) = \rho \ddot{\mathbf{u}}(\mathbf{x}, t)$$

- Using the strain-displacement relation we can write it in terms of the displacement field:

$$\nabla^T \mathbf{C} \nabla \mathbf{u}(\mathbf{x}, t) + \mathbf{f}(\mathbf{x}, t) = \rho \ddot{\mathbf{u}}(\mathbf{x}, t)$$

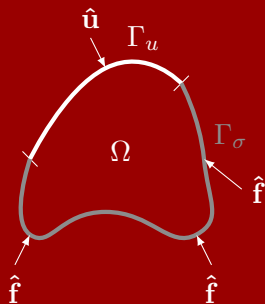
Strong form of elastodynamics

Given a deformable solid Ω with boundary Γ , and

- stiffness material matrix $\mathbf{C}$ and material density ρ ,
- prescribed body load $\mathbf{f}$ applied on Ω ,
- prescribed boundary displacement $\hat{\mathbf{u}}$ on Γ_u and surface load $\hat{\mathbf{f}}$ on Γ_σ ,
- prescribed initial (at $t = 0$) displacement $\mathbf{u}_0$ and initial velocity $\mathbf{v}_0$,

find the displacement $\mathbf{u} \in C^2(\bar{\Omega} \times [0, T], \mathbb{R}^3)$ such that

$$\begin{cases} \nabla^T \mathbf{C} \nabla \mathbf{u} + \mathbf{f} = \rho \ddot{\mathbf{u}} & \text{on } \Omega \times]0, T[\\ \mathbf{u} = \hat{\mathbf{u}} & \text{on } \Gamma_u \times]0, T[\\ \mathbf{N}^T \mathbf{C} \nabla \mathbf{u} = \hat{\mathbf{f}} & \text{on } \Gamma_\sigma \times]0, T[\\ \mathbf{u} = \mathbf{u}_0 & \text{for } t = 0 \\ \dot{\mathbf{u}} = \mathbf{v}_0 & \text{for } t = 0 \end{cases}$$



Strong and weak forms of elastodynamics

Strong form

PDE:
$$\nabla^T \mathbf{C} \nabla \mathbf{u} + \mathbf{f} = \rho \ddot{\mathbf{u}}$$

BC on Γ_u :
$$\mathbf{u} = \hat{\mathbf{u}}$$

BC on Γ_σ :
$$\mathbf{N}^T \mathbf{C} \nabla \mathbf{u} = \hat{\mathbf{f}}$$

IC at $t = 0$:
$$\mathbf{u} = \mathbf{u}_0, \dot{\mathbf{u}} = \mathbf{v}_0$$

Weak form

Virtual work:
$$\int_{\Omega} (\nabla \delta \mathbf{u})^T \mathbf{C} \nabla \mathbf{u} \, d\Omega + \int_{\Omega} \rho \delta \mathbf{u}^T \ddot{\mathbf{u}} \, d\Omega = \int_{\Gamma_\sigma} \delta \mathbf{u}^T \hat{\mathbf{f}} \, d\Gamma + \int_{\Omega} \delta \mathbf{u}^T \mathbf{f} \, d\Omega$$

Functional spaces:
$$\mathcal{U} = \{ \mathbf{u}(t, \cdot) \in H^1(\Omega) \mid \mathbf{u} = \hat{\mathbf{u}} \text{ on } \Gamma_u \}$$
$$\mathcal{V} = \{ \delta \mathbf{u} \in H^1(\Omega) \mid \delta \mathbf{u} = \mathbf{0} \text{ on } \Gamma_u \}$$

IC at $t = 0$:
$$\int_{\Omega} \rho \delta \mathbf{u}^T \mathbf{u}|_{t=0} \, d\Omega = \int_{\Omega} \rho (\delta \mathbf{u})^T \mathbf{u}_0 \, d\Omega,$$
$$\int_{\Omega} \rho \delta \mathbf{u}^T \dot{\mathbf{u}}|_{t=0} \, d\Omega = \int_{\Omega} \rho (\delta \mathbf{u})^T \mathbf{v}_0 \, d\Omega.$$

Weighted residuals

- Let $\mathbf{u}$ be a solution of

$$\nabla^T \mathbf{C} \nabla \mathbf{u} + \mathbf{f} - \rho \ddot{\mathbf{u}} = 0.$$

- Let $\mathbf{u}^h$ an approximate solution.

In general $\mathbf{u}^h$ does not satisfy the differential equation and hence results in an error or a *residual*:

$$\nabla^T \mathbf{C} \nabla \mathbf{u}^h + \mathbf{f} - \rho \ddot{\mathbf{u}}^h = \mathbf{R}^h.$$

- We impose that the residual is zero in a certain Euclidean vector space $\mathcal{V}^h$ of finite dimension. Thus

$$\langle \mathbf{R}^h, \mathbf{v}^h \rangle = 0 \quad \text{for all } \mathbf{v}^h \in \mathcal{V}^h$$

where $\langle \cdot, \cdot \rangle$ denotes the scalar product.

Discretization and Galerkin method

General ideas of Galerkin methods



- Galerkin methods are a class of numerical techniques used to transform differential equations in weak formulation, into discrete problems.
- The approximate solution is determined via a finite set of **basis functions**.
- Galerkin approximation compute the best possible approximate solution among a family of potential solutions.



Boris Galerkin 1871 - 1945

Approximate solution of the weak form

Choose subspaces $\mathcal{U}^h \subset \mathcal{U}$ and $\mathcal{V}^h \subset \mathcal{V}$ of **dimension n** and solve the *projected weak form* into such subspaces. Moreover, we impose $\mathcal{U}^h = \mathcal{V}^h$.

Displacement approximation

Let $\mathbf{u}^h(\mathbf{x}, t) = \mathbf{H}(\mathbf{x})\mathbf{q}(t)$ and $\delta\mathbf{u}^h(\mathbf{x}) = \mathbf{H}(\mathbf{x})\delta\mathbf{q}$ where

- $\mathbf{H}(\mathbf{x})$ is a $3 \times n$ matrix of l.i. **shape functions** (basis of $\mathcal{U}^h = \mathcal{V}^h$).
- $\mathbf{q}(t)$ is a $n \times 1$ vector of (*unknown*) time-dependent functions.
- $\delta\mathbf{q}$ is a $n \times 1$ vector of constants.

$$\mathbf{u}^h(\mathbf{x}, t) = \mathbf{H}(\mathbf{x})\mathbf{q}(t) \quad \text{and} \quad \delta\mathbf{u}^h(\mathbf{x}) = \mathbf{H}(\mathbf{x})\delta\mathbf{q}.$$

Formulations of elastodynamics

Strong form:

$$\nabla^T \mathbf{C} \nabla \mathbf{u} + \mathbf{f} = \rho \ddot{\mathbf{u}}$$



Weak form:

$$\int_{\Omega} (\nabla \delta \mathbf{u})^T \mathbf{C} \nabla \mathbf{u} d\Omega + \int_{\Omega} \rho \delta \mathbf{u}^T \ddot{\mathbf{u}} d\Omega = \int_{\Gamma_{\sigma}} \delta \mathbf{u}^T \hat{\mathbf{f}} d\Gamma + \int_{\Omega} \delta \mathbf{u}^T \mathbf{f} d\Omega$$



Semi-discrete weak form:

$$\mathbf{M} \ddot{\mathbf{q}}(t) + \mathbf{K} \mathbf{q}(t) = \mathbf{r}(t)$$

Definitions

- **Stiffness matrix** ($n \times n$):

$$\mathbf{K} = \int_{\Omega} \mathbf{B}^T \mathbf{C} \mathbf{B} d\Omega$$

where $\mathbf{B}$ is the $(6 \times n)$ deformation matrix defined by $\mathbf{B} = \nabla \mathbf{H}$.

- **Mass matrix** ($n \times n$):

$$\mathbf{M} = \int_{\Omega} \rho \mathbf{H}^T \mathbf{H} d\Omega.$$

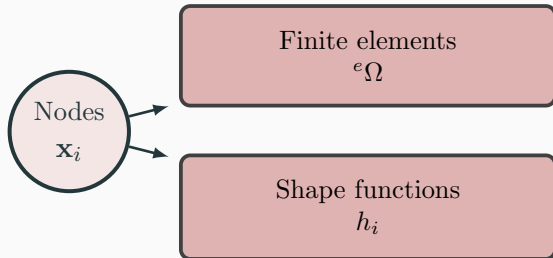
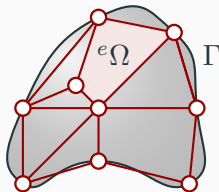
- **Applied forces vector** ($n \times 1$):

$$\mathbf{r}(t) = \int_{\Gamma_{\sigma}} \mathbf{H}^T \hat{\mathbf{f}} d\Gamma + \int_{\Omega} \mathbf{H}^T \mathbf{f} d\Omega.$$

Finite element method

Divide and conquer mentality

Finite element method: a special case of the Galerkin method, where shape functions are systematically defined in terms of nodes.



- The geometry of a structure is discretised when it is split into a mesh of finite elements $^1\Omega, \dots, ^m\Omega$ (smaller components or regular shapes).
- Each shape function is non-zero only on a very limited number of finite elements (*compact support*).

Displacements approximation in finite element method

Let p the number of nodes of the mesh.

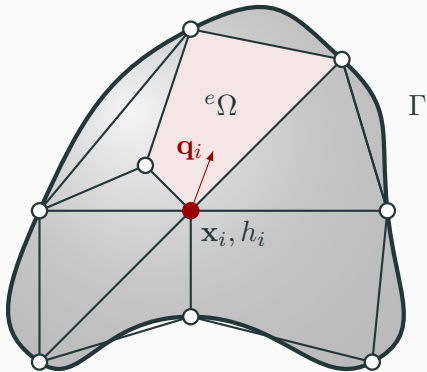
$$\mathbf{u}^h(\mathbf{x}, t) = \mathbf{H}(\mathbf{x})\mathbf{q}(t) = \sum_{i=1}^p h_i(\mathbf{x})\mathbf{q}_i(t)$$

■ $\mathbf{H}(\mathbf{x})$ is a $3 \times 3p$ matrix of **shape functions**:

$$\mathbf{H} = [h_1\mathbf{I} \mid h_2\mathbf{I} \mid \dots \mid h_i\mathbf{I} \mid \dots \mid h_p\mathbf{I}]$$

$\mathbf{I}$ is the 3×3 identity matrix.

■ $\mathbf{q}(t)$ is a $3p \times 1$ vector of (*unknown*) nodal displacements.



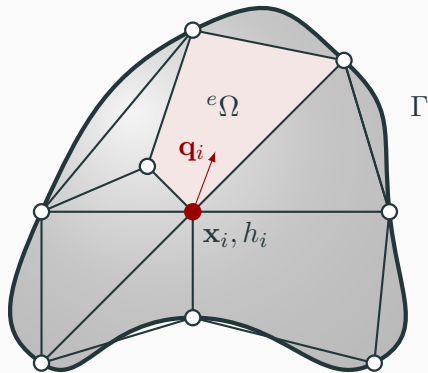
Global nodal shape functions requirements

Properties of h_i :

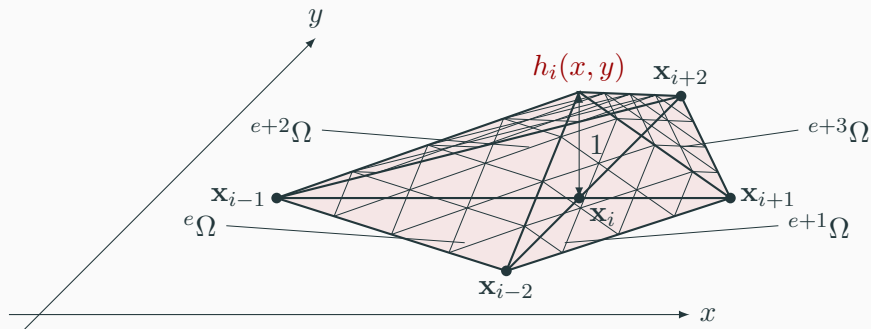
- Linearly independent polynomial basis.
- Satisfy Kronecker delta property:

$$h_i(\mathbf{x}_i) = 1 \quad \text{and} \quad h_i(\mathbf{x}_j) = 0.$$

- Vanish on non-adjacent elements.
- Continuous at interfaces.
- Differentiable inside elements.
- Ensure rigid body motion & constant deformations.



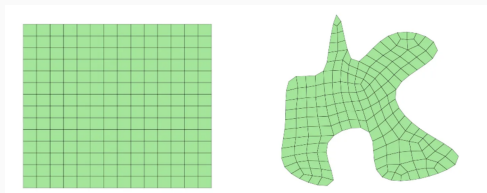
Bilinear triangular shape functions



Drawbacks of the global approach of FEM

- ✗ Limited capability in handling complex (unstructured) mesh topologies.
- ✗ Computationally expensive: *it requires defining one shape function per node !*
- ✗ Limited utilization of the compact support of nodal shape functions.

Local approach: provides a quicker and more systematic way to compute the stiffness and mass matrices and the applied forces vector:



(Credit: Onscale - structured vs unstructured meshes)

Local approach of the finite element method

Physical structure:

Ω



Elements:

$^1\Omega, ^2\Omega, \dots, ^m\Omega$

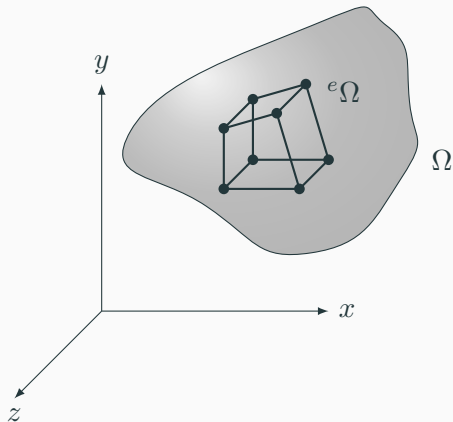


Master elements:

$\Omega^a, \Omega^b, \dots$



Localization

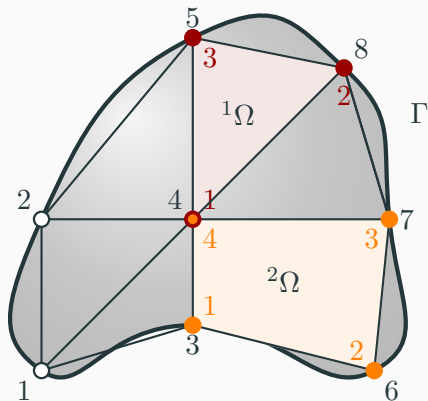


- Let p be the number of nodes.
- Let m be the number of finite elements in the mesh.
- Let ${}^e\Omega$ a generic finite element.
- Let ep the number of nodes in the generic element ${}^e\Omega$.

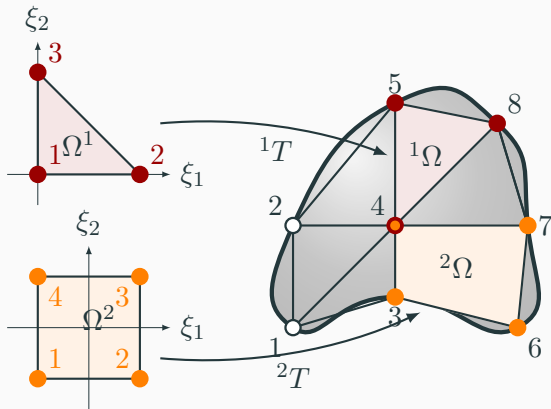
The local finite element point of view (in 2d)

Every finite element ${}^e\Omega$ has:

- Number of nodes ep ,
- Connectivity (${}^ep \times 1$),
- ${}^e\mathbf{q}$ vector of nodal displacements ($2{}^ep \times 1$),
- ${}^e\mathbf{u}$ vector of displacements (2×1),
- eh_i local shape functions ($i = 1, \dots, {}^ep$),
- ${}^e\mathbf{H}$ local shape functions matrix ($2 \times 2{}^ep$),
- ${}^e\mathbf{K}$ local stiffness matrix ($2{}^ep \times 2{}^ep$),
- ${}^e\mathbf{M}$ local mass matrices ($2{}^ep \times 2{}^ep$),
- ${}^e\mathbf{r}$ local loads vector ($2{}^ep \times 1$).



Paradigm shift: many with few



👉 **Main idea:** Local shape functions $^e h_i$ are deduced from a **small set of archetypal shape functions** $^a h_i$ defined on an archetypal element.

👉 **Clever idea:** the transformation $^e T$ itself is defined in terms of archetypal shape functions and nodes coordinates:

$$^e T : \mathbf{x}(\boldsymbol{\xi}) = {}^a \mathbf{H}(\boldsymbol{\xi}) {}^e \mathbf{x} = \sum_{i=1}^{e_p} {}^a h_i(\boldsymbol{\xi}) {}^e \mathbf{x}_i$$

Concrete example: bilinear triangular element (2d)

Node id local	Local Coord	Node id global	Global coord.
1	(0,0)	4	(x_4, y_4)
2	(1,0)	8	(x_8, y_8)
3	(0,1)	5	(x_5, y_5)

Base (archetypal) functions:

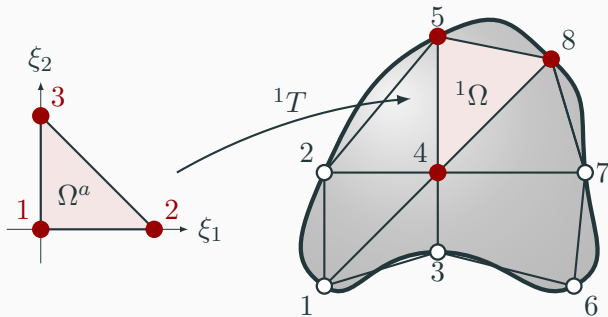
$$^a h_1 = 1 - \xi_1 - \xi_2$$

$$^a h_2 = \xi_1$$

$$^a h_3 = \xi_2$$

Transformation:

$$^1 T : \begin{cases} x(\xi_1, \xi_2) = x_4 h_1 + x_8 h_2 + x_5 h_3 = x_4 + (x_8 - x_4)\xi_1 + (x_5 - x_4)\xi_2 \\ y(\xi_1, \xi_2) = y_4 h_1 + y_8 h_2 + y_5 h_3 = y_4 + (y_8 - y_4)\xi_1 + (y_5 - y_4)\xi_2 \end{cases}$$



Consequence: integration over archetypal elements

$$\begin{aligned}\int_{{}^e\Omega} (\dots) d\Omega &\xrightarrow{\text{replaced by}} \int_{{}^e\Omega^a} (\dots) {}^e j d\xi_1 d\xi_2 \\ \frac{\partial}{\partial x_i} &\xrightarrow{\text{replaced by}} \frac{\partial \xi_1}{\partial x_i} \partial_{\xi_1} + \frac{\partial \xi_2}{\partial x_i} \partial_{\xi_2}\end{aligned}$$

where ${}^e j$ is the determinant of the Jacobian matrix ${}^e \mathbf{J}$ of the transformation ${}^e T$.

 For each transformation, it is necessary to compute the Jacobian matrix, its determinant, and its inverse.

Concrete example: local mass and stiffness matrices

■ Local mass matrix:

$$\begin{aligned} {}^1\mathbf{M} &= \int_{\Omega^a} \rho {}^a\mathbf{H}^T {}^a\mathbf{H} {}^1j \, d\xi \\ &= \int_{\Omega^a} \rho \begin{bmatrix} {}^ah_1^2\mathbf{I} & {}^ah_1{}^ah_2\mathbf{I} & {}^ah_1{}^ah_3\mathbf{I} \\ {}^ah_2{}^ah_1\mathbf{I} & {}^ah_2^2\mathbf{I} & {}^ah_2{}^ah_3\mathbf{I} \\ {}^ah_3{}^ah_1\mathbf{I} & {}^ah_3{}^ah_2\mathbf{I} & {}^ah_3^2\mathbf{I} \end{bmatrix} {}^1j \, d\xi_2 d\xi_1 \end{aligned}$$

■ Local stiffness matrix:

$${}^1\mathbf{K} = \int_{\Omega^a} \begin{bmatrix} \cdots & (\nabla_{\xi} {}^ah_i {}^1\mathbf{J}^{-1})^T \mathbf{C} (\nabla_{\xi} {}^ah_j {}^1\mathbf{J}^{-1}) & \cdots \\ \cdots & & \cdots \end{bmatrix} {}^1j \, d\xi_2 d\xi_1$$

Assembly of the mass matrix: local to global

$${}^1\mathbf{M} = \int_{\Omega^a} \rho \begin{bmatrix} {}^a h_1^2 \mathbf{I} & {}^a h_1 {}^a h_2 \mathbf{I} & {}^a h_1 {}^a h_3 \mathbf{I} \\ {}^a h_2 {}^a h_1 \mathbf{I} & {}^a h_2^2 \mathbf{I} & {}^a h_2 {}^a h_3 \mathbf{I} \\ {}^a h_3 {}^a h_1 \mathbf{I} & {}^a h_3 {}^a h_2 \mathbf{I} & {}^a h_3^2 \mathbf{I} \end{bmatrix} {}^e j d\xi_2 d\xi_1 = \begin{bmatrix} {}^1\mathbf{M}_{11} & {}^1\mathbf{M}_{12} & {}^1\mathbf{M}_{13} \\ {}^1\mathbf{M}_{21} & {}^1\mathbf{M}_{22} & {}^1\mathbf{M}_{23} \\ {}^1\mathbf{M}_{31} & {}^1\mathbf{M}_{32} & {}^1\mathbf{M}_{33} \end{bmatrix}$$

We assembly local mass matrices into the global ones using the assembly operator:

$$\mathbf{M} = \bigcup_{e=1}^m {}^e\mathbf{M} = \sum_{e=1}^m {}^e\mathbf{L}^T {}^e\mathbf{M} {}^e\mathbf{L}$$

where

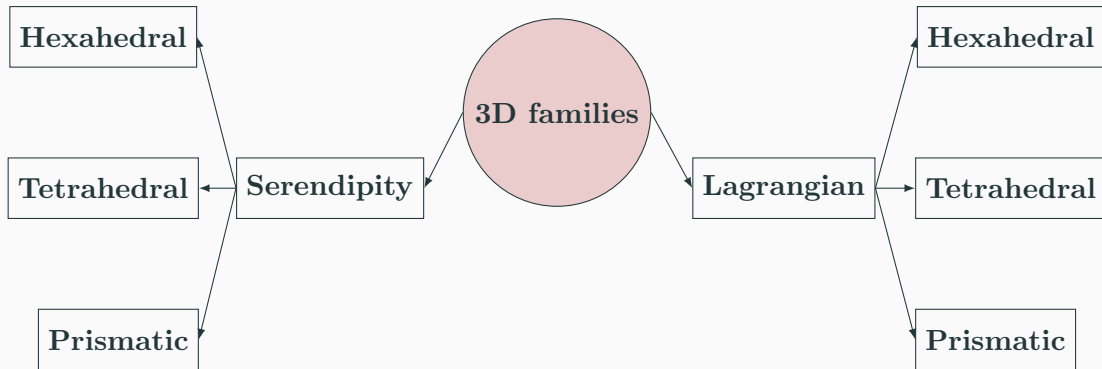
$${}^1\mathbf{L} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}$$

Node id. local	Node id. global
1	4
2	8
3	5

Assembly of the mass matrix: local to global

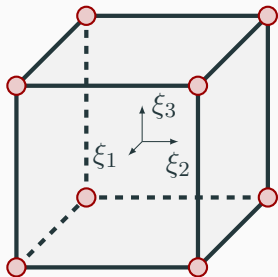
$${}^1\mathbf{L}^T {}^1\mathbf{M} {}^1\mathbf{L} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & {}^1\mathbf{M}_{11} & {}^1\mathbf{M}_{13} & 0 & 0 & {}^1\mathbf{M}_{12} \\ 0 & 0 & 0 & {}^1\mathbf{M}_{31} & {}^1\mathbf{M}_{33} & 0 & 0 & {}^1\mathbf{M}_{32} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & {}^1\mathbf{M}_{21} & {}^1\mathbf{M}_{23} & 0 & 0 & {}^1\mathbf{M}_{22} \end{bmatrix}$$

Three-dimensional families of finite elements

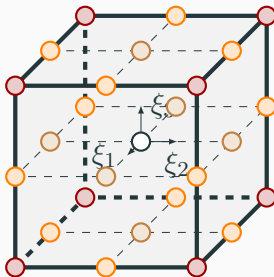


Hexahedral elements

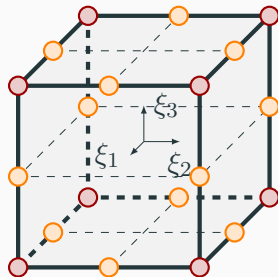
Trilinear
Lagrangian



Triquadratic
Lagrangian

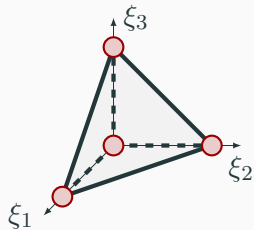


Triquadratic
serendipity

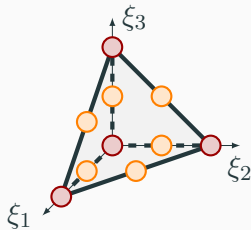


Tetrahedral elements

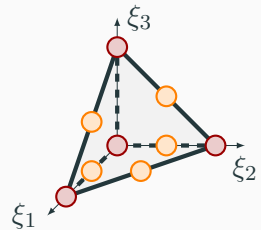
Trilinear
Lagrangian



Triquadratic
Lagrangian

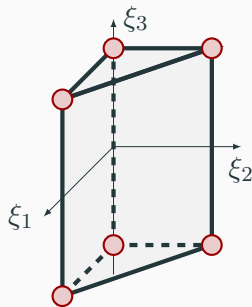


Triquadratic
serendipity

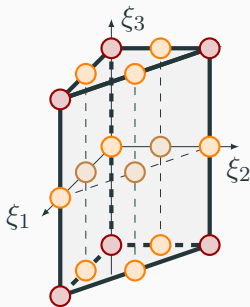


Prismatic elements

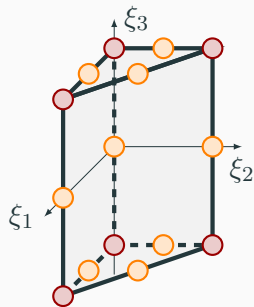
Trilinear
Lagrangian



Triquadratic
Lagrangian



Triquadratic
serendipity



Gauss-Legendre quadrature

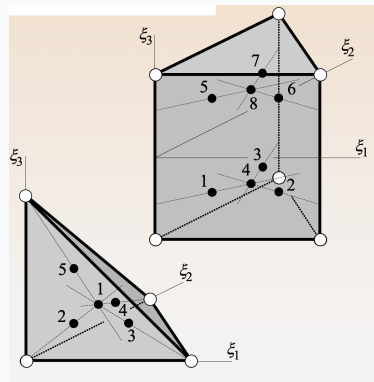
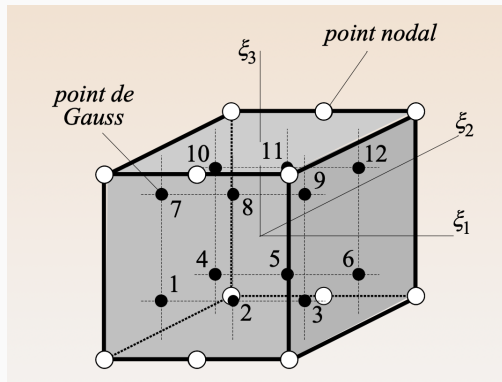
- Numerical integration helps automate the calculation of the elementary quantities ${}^e\mathbf{K}$, ${}^e\mathbf{M}$ and ${}^e\mathbf{r}$.
- Integration is approximated as:

$$\int_{\Omega^a} f(\boldsymbol{\xi}) d\boldsymbol{\xi} \approx \sum_{i=1}^{r_1} \sum_{j=1}^{r_2} \sum_{k=1}^{r_3} \omega_i^1 \omega_j^2 \omega_k^3 f(\xi_1^i, \xi_2^j, \xi_3^k)$$

where ξ_j^i are known as *Gauss points* and ω_i^j are their associated *Gauss weights*.

👉 If f is a polynomial of degree $d_i = 2r_i - 1$ in the variable ξ_i , then the Gauss-Legendre approximation formula is **exact**.

An illustration of Gauss points - 3d elements

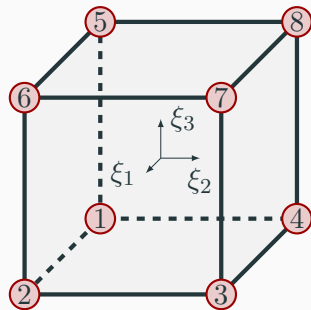


(Credit: Thomas Gmür - Dynamique numérique des solides et des structures)

Number of Gauss points for exact integration

Trilinear hexahedral element

- Basis functions are linear in each variable.
- Exact integration with:
 - $2 \times 2 \times 2$ Gauss points for ${}^e\mathbf{M}$
 - $1 \times 1 \times 1$ Gauss points for ${}^e\mathbf{K}$if the element is not distorted.
- Note that in case of distortion:
 - Jacobian is a function of $\boldsymbol{\xi}$: ${}^e j = {}^e j(\boldsymbol{\xi})$
 - ${}^e \mathbf{J}^{-1} = {}^e \mathbf{J}^{-1}(\boldsymbol{\xi})$ function of $\boldsymbol{\xi}$ at the denominator.
 - $\frac{\partial {}^e \mathbf{H}}{\partial \mathbf{x}} = \frac{\partial {}^a \mathbf{H}}{\partial \boldsymbol{\xi}} {}^e \mathbf{J}^{-1}$ is not a polynomial function of $\boldsymbol{\xi}$ anymore.



👉 Gauss quadrature will never be exact in case of distortion.

Types of structural analysis

Static, modal, and transient analysis

- **Static analysis:** determines deformations $\mathbf{q}$ due to constant loads.

$$\mathbf{K}\mathbf{q} = \mathbf{r}$$

- **Modal analysis:** studies dynamic properties in the frequency domain.

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{0}$$

Natural frequency: key structural property essential in structural design.

- Lower frequencies $\rightarrow$ higher displacement amplitudes $\rightarrow$ more dangerous.
- Resonance when external excitation frequency matches a natural frequency.
- Prolonged resonance $\rightarrow$ catastrophic failure.

- **Transient analysis:** examines time-dependent structural responses $\mathbf{q}(t)$ to time-dependent excitations $\mathbf{r}(t)$.

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{r}(t)$$

Modal analysis overview

Free undamped
discrete vibration:

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{0}$$



Ansatz:

$$\mathbf{q}(t) = \alpha \mathbf{p} \cos(\omega t + \varphi)$$



Generalized
eigenvalue problem:

$$(\mathbf{K} - \omega^2 \mathbf{M})\mathbf{p} = \mathbf{0}$$

Solving the eigenvalue problem:

$$n = \dim(\mathbf{K}) = \dim(\mathbf{M})$$

Eigenvalues (*natural frequencies squared*):
 $\lambda_j = \omega_j^2$ are the roots of the characteristic polynomial:

$$\det(\mathbf{K} - \lambda \mathbf{M}) = 0.$$

Eigenvectors (*modal shapes*): $\mathbf{p}_j$ are the solution of the equation

$$(\mathbf{K} - \lambda_j \mathbf{M})\mathbf{p}_j = \mathbf{0}.$$

A priori error estimates for eigenvalues and eigenvectors

$$\lambda_i^e \leq \lambda_i \leq \lambda_i^e + c h^{2(m-k+1)} (\lambda_i^e)^{m+1}$$
$$\|\mathbf{p}_i - \mathbf{p}_i^e\|_0 \leq c h^{\min(m+1, 2(m-k+1))} (\lambda_i^e)^{(m+1)/2}$$

- λ_i are the *approximated* eigenvalues and $\mathbf{p}_i$ the corresponding eigenvectors,
- λ_i^e are the *exact* eigenvalues and $\mathbf{p}_i^e$ the corresponding exact eigenvectors,
- h represents the characteristic mesh size,
- m is the degree of the highest complete polynomial used,
- k is the the highest derivative order appearing in the weak form,
- c is a constant independent of h ,
- $\|\cdot\|_0$ Euclidean L^2 norm.

Rigid body modes

In the semi-discrete weak form obtained via finite element discretization:

- The **mass matrix** $\mathbf{M}$ is symmetric and strictly positive definite.
- The **stiffness matrix** $\mathbf{K}$ is symmetric and positive semi-definite:

$$\mathbf{K}\mathbf{p} = \mathbf{0} \quad \text{for certain nonzero vectors } \mathbf{p}.$$

Consequently, the eigenvalues λ_j of the generalized eigenvalue problem are all real and non-negative:

$$0 \leq \omega_1 \leq \omega_2 \leq \cdots \leq \omega_n.$$

Rigid body modes: zero eigenvalues (i.e., $\omega_j = 0$) correspond to *rigid body motions*, where the system undergoes displacement without internal deformation.

Orthonormalization of mode shapes

Let $\mathbf{p}_i$ and $\mathbf{p}_j$ two eigenvectors corresponding to the eigenvalues λ_i and λ_j , then

$$\mathbf{p}_i^T \mathbf{M} \mathbf{p}_j = \delta_{ij} \quad \text{and} \quad \mathbf{p}_i^T \mathbf{K} \mathbf{p}_j = \omega_i^2 \delta_{ij}$$

where δ_{ij} represent Kronecker symbol.

Consequences: if we organize the modal vectors $\mathbf{p}_i$ in a so-called modal matrix $\mathbf{P}$:

$$\mathbf{P} = \left[\begin{array}{c|c|c|c} \mathbf{p}_1 & \mathbf{p}_2 & \dots & \mathbf{p}_n \end{array} \right]$$

then

$$\mathbf{P}^T \mathbf{M} \mathbf{P} = \mathbf{I} \quad \text{and} \quad \mathbf{P}^T \mathbf{K} \mathbf{P} = \mathbf{\Lambda}$$

where $\mathbf{I}$ is the identity matrix of order n and $\mathbf{\Lambda}$ the *spectral matrix*:

$$\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n) = \text{diag}(\omega_1^2, \dots, \omega_n^2).$$

Small and large scale problems

Economical computation algorithms for extracting $(\lambda_j, \mathbf{p}_j)$ for $1 \leq j \leq n_{modes}$ are required in practical applications, where $n_{modes} \ll n$ is the number of desired eigenpairs.

Small-scale problems

- system matrices are of modest size $n \leq 250$ or $250 \leq n \leq 2500$ and small bandwidth matrices.
- An explicit reduction to standard eigenvalue form is typically employed:
 - *Generalized Jacobi method*
 - *Householder QR-algorithm*

Large-scale problems

- $n \geq 2500$.
- The most effective algorithms are:
 - *Subspace iteration method*,
 - *Lanczos' method*,
 - *Irons-Guyan reduction method*,
 - *Arnoldi method*
 - *Davidson method*.

Rayleigh's quotient

- Rayleigh's quotient of a generic vector $\mathbf{w}$:

$$\mathcal{R}(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{K} \mathbf{w}}{\mathbf{w}^T \mathbf{M} \mathbf{w}}$$

- For a given eigenvector $\mathbf{p}_i$, the Rayleigh quotient provides the corresponding eigenvalue ω_i^2 :

$$\mathcal{R}(\mathbf{p}_i) = \lambda_i = \omega_i^2.$$

- The first variation of Rayleigh's quotient is zero at the eigenvectors:

$$\delta \mathcal{R}(\mathbf{w}) = 0 \quad \text{if and only if} \quad \mathbf{w} = \mathbf{p}_i,$$

where $\mathbf{p}_i$ is an eigenvector.

- **Bounding theorem:** $\lambda_1 \leq \mathcal{R}(\mathbf{w}) \leq \lambda_n$

The Rayleigh quotient can be used to approximate the smallest eigenvalue $\lambda_1 = \omega_1^2$ by minimizing $\mathcal{R}(\mathbf{w})$ over all non-zero vectors $\mathbf{w}$.

General idea of inverse and forward iteration

Goal: compute **one** eigenvector of the generalized eigenproblem $\mathbf{Kp} = \lambda \mathbf{Mp}$.

Idea: Use an *iterative* improvement of an assumed eigenvector.

- Start from an arbitrary vector $\mathbf{p}^{(0)}$.
- Generate a sequence of vectors $\mathbf{p}^{(1)}, \mathbf{p}^{(2)}, \dots$ that converge to an eigenvector.
- Compute an approximate eigenvalue using Rayleigh's quotient:

$$\lambda^{(k)} = \mathcal{R}(\mathbf{p}^{(k)}).$$

Inverse iteration:

- Form a “force” vector: $\mathbf{R}_1 = \mathbf{Mp}^{(0)}$.
- Solve the static-like equilibrium problem:
$$\mathbf{Kp}^{(1)} = \mathbf{R}_1,$$
- Repeat until convergence is met.

Forward iteration:

- Compute $\mathbf{R}_1 = \mathbf{Kp}^{(0)}$.
- Solve
$$\mathbf{Mp}^{(1)} = \mathbf{R}_1,$$
- Repeat until convergence is met.

Inverse iteration algorithm

Goal: Compute **one** eigenvector using iterative refinement.

Algorithm:

- ① Choose initial guess $\mathbf{p}^{(0)}$, spectral shift σ , tolerance ε .
- ② Define shifted stiffness: $\mathbf{K}_\sigma = \mathbf{K} + \sigma\mathbf{M}$
- ③ For $k = 1, 2, \dots$ until convergence:
 - ① Solve: $\mathbf{K}_\sigma \bar{\mathbf{p}}^{(k)} = \mathbf{M} \mathbf{p}^{(k-1)}$
 - ② Normalize: $\mathbf{p}^{(k)} = \bar{\mathbf{p}}^{(k)} / \sqrt{(\bar{\mathbf{p}}^{(k)})^T \mathbf{M} \bar{\mathbf{p}}^{(k)}}$
 - ③ Check: $\|\mathbf{p}^{(k)} - \mathbf{p}^{(k-1)}\| < \varepsilon$
- ④ Output: eigenvector $\mathbf{p}^{(k)}$, eigenvalue $\lambda^{(k)} = \mathcal{R}(\mathbf{p}^{(k)})$

Key points:

- Converges to eigenvector closest to σ if $\mathbf{p}^{(0)}$ not $\mathbf{M}$ -orthogonal to it.
- Without shift ($\sigma = 0$) $\rightarrow$ eigenvector of smallest eigenvalue.
- Convergence rate: $\rho = \frac{|\lambda_j - \sigma|}{|\lambda_{j+1} - \sigma|}$

Inverse iteration on q vectors

Suppose that we perform a simultaneous inverse iteration on q eigenvectors:

$$\mathbf{P}^{(k)} = (\mathbf{K}^{-1}\mathbf{M})\mathbf{P}^{(k-1)} = \dots = (\mathbf{K}^{-1}\mathbf{M})^k \mathbf{P}^{(0)}$$

Define the subspace $\mathcal{S}^{(k)}$ of rank q , spanned by the vectors $\{\mathbf{p}_i^{(k)}\}$.

$\mathbf{P}^{(k)} = [\mathbf{p}_1^{(k)}, \dots, \mathbf{p}_q^{(k)}]$ forms a *non-orthogonal* basis of vector spaces $\mathcal{S}^{(k)}$.

- ✗ All columns of $\mathbf{P}^{(k)}$ tend toward $\mathbf{p}_1$
- ✗ Collinearity if no orthogonalization is applied !

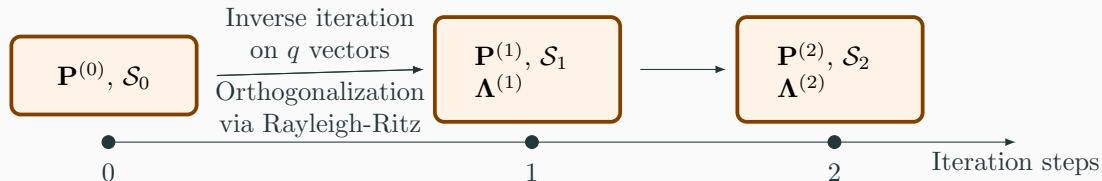
Orthogonalization of vectors $\mathbf{p}_i^{(k)}$ at each iteration needed !

Subspace iteration method

Goal: Compute the lowest $n_{modes} \ll n$ eigenpairs $(\mathbf{p}_i, \lambda_i)$ of using a q -dimensional subspace iteration.

Key ideas and advantages:

- Iteration is performed on a subspace $\mathcal{S}_k = \text{span}(\mathbf{p}_1^{(k)}, \dots, \mathbf{p}_q^{(k)})$, not individual vectors.
- At each iteration, inverse iteration is applied to all q vectors in the subspace and then orthogonalized via Rayleigh-Ritz.
- Easier to initialize a subspace close to the span of required eigenvectors than individual vectors.



Subspace iteration algorithm

Inputs: $\mathbf{K}, \mathbf{M}, n_{modes}, q > n_{modes}, \mathbf{P}^{(0)} \in \mathbb{R}^{n \times q}, \sigma, \varepsilon$

Iteration: For $k = 1, 2, \dots$ until convergence:

- ① Solve $\mathbf{K}_\sigma \overline{\mathbf{P}^{(k)}} = \mathbf{M} \mathbf{P}^{(k-1)}$ (inverse iteration on q vectors)
- ② Project and solve small eigenproblem:

$$\mathbf{K}^{(k)} = (\overline{\mathbf{P}^{(k)}})^T \mathbf{K} \overline{\mathbf{P}^{(k)}}, \quad \mathbf{M}^{(k)} = (\overline{\mathbf{P}^{(k)}})^T \mathbf{M} \overline{\mathbf{P}^{(k)}}$$

$$\mathbf{K}^{(k)} \mathbf{Z}^{(k)} = \mathbf{M}^{(k)} \mathbf{Z}^{(k)} \mathbf{\Lambda}^{(k)}$$

- ③ Update modal matrix: $\mathbf{P}^{(k)} = \overline{\mathbf{P}^{(k)}} \mathbf{Z}^{(k)}$, ensure $(\mathbf{P}^{(k)})^T \mathbf{M} \mathbf{P}^{(k)} = \mathbf{I}$
- ④ Check convergence: $\frac{|\lambda_i^{(k)} - \lambda_i^{(k-1)}|}{\lambda_i^{(k)}} < \varepsilon$

Notes:

- Orthogonalization is critical to prevent collinearity of iteration vectors.
- Starting vectors should be linearly independent; larger q increases convergence rate but also computational cost.
- After convergence, use Sturm sequence to verify the number of eigenvalues computed.

Starting iteration vectors $\mathbf{P}^{(0)}$

- The starting iteration vectors in $\mathbf{P}^{(0)}$ should not be too close to each other, i.e., they should be linearly independent.
- Higher convergence rate can be obtained by increasing q . However, using more iteration vectors will also increase the computer effort for one iteration. In practice

$$q = \min(2n_{modes}, n_{modes} + 8)$$

Empirical guidelines

- The first column of $\mathbf{P}^{(0)}$ should be the diagonal of $\mathbf{M}$. This ensures that all mass degrees of freedom are excited.
- The other columns in $\mathbf{P}^{(0)}$, except for the last column, should be unit vectors $\mathbf{e}_i$, with entries +1 at the degrees of freedom with the smallest ratios k_{ii}/m_{ii} , and the last column in $\mathbf{P}^{(0)}$ should be a random vector with values in $[0, 1]$.

Convergence of the subspace iteration method

- Iteration is performed with q vectors, $q > n_{modes}$, but convergence is measured only on the approximations obtained for the n_{modes} smallest eigenvalues.
- If the starting iteration vectors in $\mathbf{P}^{(0)}$, are not orthogonal to any one of the required eigenvectors $\mathbf{p}_1, \dots, \mathbf{p}_{n_{modes}}$, then as $k \rightarrow \infty$

$$\tilde{\mathcal{S}}_k = \text{span}(\mathbf{p}_1^{(k)}, \dots, \mathbf{p}_{n_{modes}}^{(k)}) \rightarrow \mathcal{S}_\infty = \text{span}(\mathbf{p}_1, \dots, \mathbf{p}_{n_{modes}})$$

- Assuming that in the iteration the vectors in $\mathbf{P}^{(k+1)}$ are ordered in such way that the i -th diagonal element in $\mathbf{\Lambda}^{(k+1)}$ is larger than the $(i-1)$ -st element for $i = 2, \dots, p$, then the i -th column in $\mathbf{P}^{(k+1)}$ converges to $\mathbf{p}_i$ and the convergence rate is

$$\lambda_i / \lambda_{p+1}$$

Although this is an asymptotic convergence rate, it indicates that the smallest eigenvalues converge fastest.

Final Sturm sequence check

After convergence is achieved, a final check based on the Sturm sequence property can be performed to ensure that the number of computed eigenvalues below a certain value is correct.

- The Sturm sequence property states that the number of eigenvalues of the generalized eigenvalue problem $\mathbf{K}\mathbf{p} = \lambda\mathbf{M}\mathbf{p}$ that are less than a given value λ^* is equal to the number of negative pivots obtained when performing an $\mathbf{LDL}^T$ factorization of the matrix $\mathbf{K} - \lambda^*\mathbf{M}$.
- This property can be used to verify that the number of computed eigenvalues below a certain threshold matches the expected count, providing an additional layer of validation for the results obtained from the subspace iteration method.